

CEG 4424/6424 (Security Attacks and Defenses) Homework #3

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Question 1. (5 points) Let $Z_n^* = \{x \mid 1 \leq x \leq n-1, \text{ where } \text{GCD}(x, n)=1\}$. $Z_{12}^* = ?$

Answer: {1, 5, 7, 11}

Question 2. (5 points) Find integer x such that $3 \cdot x = 1 \pmod{23}$.

Answer: $x = 8$ because $3 \cdot 8 = 24$ and $24 \pmod{23} = 1$.

Question 3 (10 points). Consider the RSA algorithm. Let the two prime numbers, $p=11$ and $q=31$. You need to derive appropriate public key (e,n) and private key (d,n) .

(a) (2 points) Can we pick $e=5$? If yes, what will be the corresponding (d,n) ?

Answer: first, calculate n and $\phi(n)$: $n = p \cdot q = 11 \cdot 31 = 341$

$$\phi(n) = (p-1)(q-1) = 10 \cdot 30 = 300$$

For e to be valid, it must be coprime with $\phi(n)$ (i.e., $\text{GCD}(5, 300) = 1$). But because 300 is divisible by 5, $\text{GCD}(5, 300) = 5$, not 1.

Therefore, $e=5$ is not a valid public key and we cannot pick it.

(b) (5 points) Can we pick $e=17$? If yes, what will be the corresponding (d,n) ?
Calculation Reference is given as follows.

$17 \cdot 51 = 867$
$17 \cdot 52 = 884$
$17 \cdot 53 = 901$
$17 \cdot 54 = 918$

Answer: $\text{GCD}(17, 300) = 1$, so $e=17$ is valid. We need to find d , the modular multiplicative inverse of $e \pmod{\phi(n)}$, such that $(d \cdot e) \pmod{\phi(n)} = 1$.

Using the provided reference calculations, we are looking for a multiple of 17 that is $1 \pmod{300}$:

$$17 \cdot 53 = 901$$

$$901 \pmod{300} = 901 - (3 \cdot 300) = 901 - 900 = 1$$

Therefore, the corresponding $(d, n) = (53, 341)$.

(c) (3 points) Use $e=17$, how to encrypt the number 3? You do not need to provide the encrypted value.

Answer: Encryption is performed using the formula: $\text{Ciphertext} = (\text{Message})^e \pmod{n}$. To encrypt the number 3, we compute $3^{17} \pmod{341}$.

Question 4 (10 points, Question for Graduate Students) Is $PU=(23, 55)$ and $PR=(7,55)$ a valid RSA public key and private key pair? Show your work. (Hint: use the RSA key generation algorithm).

Answer: first, we factor n to find p and q .

$n = 55$. The prime factors of 55 are 5 and 11.

So, $p = 5$, $q = 11$.

Calculate $\phi(n)$:

$$\phi(n) = (p-1)(q-1) = (4)(10) = 40.$$

for the keys to be valid, the product of the public exponent ($e=23$) and the private exponent ($d=7$) must be congruent to 1 modulo $\phi(n)$.

we calculate $e * d = 23 * 7 = 161$.

Now computing $161 \bmod 40$: $161 / 40 = 4$ with a remainder of 1 (since $4 * 40 = 160$, and $161 - 160 = 1$). And because $(e * d) \bmod \phi(n) = 1$, this is a valid RSA key pair.

Question 5 (20 points). The Common Weakness Enumeration (CWE) is a category system for hardware and software weaknesses and vulnerabilities. Please describe what is CWE-322, CWE-328, CWE-336, and CWE-1240. (Hint: Please go to the website cwe.mitre.org to find the descriptions of these CWEs. You should describe the content in your own words by rephrasing the sentences from the online resources.)

Answer: The description of the CWEs are as follows:

- **CWE-322: Key Exchange without Entity Authentication**
This weakness occurs when a product performs a key exchange with another party but fails to verify the identity of that party before establishing the secure communication channel. By skipping or improperly implementing authentication, the product cannot be sure who it is exchanging keys with. This allows an attacker to impersonate a trusted entity, potentially leading to a man-in-the-middle attack where they can intercept, decrypt, or modify the supposedly secure traffic.
- **CWE-328: Use of Weak Hash**
This weakness refers to the use of a hashing algorithm that is no longer considered secure for cryptographic purposes. A weak hash is one where an attacker can reasonably perform a preimage attack (find the original input), a 2nd preimage attack (find a different input that produces the same hash), or a birthday attack (find two different inputs that hash to the same value) more efficiently than brute force. This compromises the integrity of any system relying on the hash, such as password storage or digital signatures. Common examples of algorithms now considered weak include MD5 and SHA-1.
- **CWE-336: Same Seed in Pseudo-Random Number Generator (PRNG)**
This describes a situation where a Pseudo-Random Number Generator (PRNG) is initialized with the same seed value every time the product starts up or a function is called. Since PRNGs are deterministic algorithms, using the same seed will generate the exact same sequence of "random" numbers each time. This predictability allows an attacker, if they know or can guess the seed, to determine all future "random" values, completely breaking the security of any mechanism that depends on randomness (e.g., generating session IDs, cryptographic keys).
- **CWE-1240: Use of a Cryptographic Primitive with a Risky Implementation**

This weakness happens when a system needs encryption or hashing but uses a cryptographic algorithm in an unsafe or non-standard way. It is not only about choosing a weak algorithm; it also includes implementing a strong algorithm incorrectly. For example, developers might create their own version of a cipher, modify a standard algorithm without proper analysis, or build it into hardware with design flaws. Even if the underlying algorithm is secure, a poor or custom implementation can introduce serious vulnerabilities. These risky approaches are dangerous because they have not been thoroughly tested, reviewed, or analyzed like well-established cryptographic implementations.